

A Norming Family of Uniform Numerical Bollobas Operators

Run fa_banach_001, agent agent_lane_04

June 18, 2026

Source and Target

The source paper is S. Dantas, M. Jung, and O. Roldan, *Norm attaining operators which satisfy a Bollobas type theorem*, arXiv:1910.05726. After Proposition 2.8, the authors construct compact operators $T \in \mathcal{A}_\nu(H)$, where H is a separable infinite-dimensional real Hilbert space, such that

$$1 = \nu(T) < \|T\|,$$

and such that the Bollobas modulus in the definition of $\mathcal{A}_\nu(H)$ can be chosen independently of the particular operator T . They write:

Let us recall that in Corollary 2.6, we proved that if a compact operator T defined on a Hilbert space is such that $\nu(T) = \|T\| = 1$, then T must belong to the set $\mathcal{A}_{\text{nu}}(H)$. However, the following result (inspired by [2, Example 1.9]) provides us a wide class of operators $T \in \mathcal{A}_{\text{nu}}(H)$ such that $1 = \nu(T) < \|T\|$ and, in particular, examples of operators which belong to $\mathcal{A}_{\text{nu}}$ but not to $\mathcal{A}_{\|\cdot\|}$. Notice, by item (iii) below, that T belongs to the set $\mathcal{A}_{\text{nu}}$ in a uniform sense, that is, the η does not depend on the operator T defined there. We do not know how often this happens, that is, we do not know, for instance, whether the set of such an operators could be norming for the whole space.

Proposition 2.8. *Let H be a separable infinite dimensional real Hilbert space. Then, there is $T \in \mathcal{L}(H)$ such that*

- (i) *T is a compact operator.*
- (ii) *$1 = \nu(T) < \|T\|$ and T attains its numerical radius.*
- (iii) *given $\varepsilon > 0$, there is $\eta(\varepsilon) > 0$ such that whenever $x_0 \in S_H$ satisfies*

$$|\langle Tx_0, x_0 \rangle| > 1 - \eta(\varepsilon),$$

there is $x_1 \in S_H$ such that $\nu(T) = \langle Tx_1, x_1 \rangle = 1$ and $\|x_1 - x_0\| < \varepsilon$.

In particular, $T \in \mathcal{A}_{\text{nu}}(H)$ and $T \notin \mathcal{A}_{\|\cdot\|}(H, H)$.

Proof. Let $0 < \alpha \leq 1$ and $\{\alpha_n\}$ be a sequence such that $|\alpha_1| > 1$, $-1 < \alpha_n < 1$ for $n \geq 2$, and $\alpha_n \rightarrow 0$ as $n \rightarrow \infty$. Let $\{J_1, J_2, J_3\}$ be a partition of $\mathbb{N}$ such that $|J_1| = |J_2| = \aleph_0$,

The phrase “norming for the whole space” is informal in the source. We answer the natural numerical-radius interpretation: whether such operators can form a family that norms vectors of H through their quadratic forms.

Definitions

For an operator $T \in \mathcal{L}(H)$, its numerical radius is

$$\nu(T) = \sup_{\|x\|=1} |\langle Tx, x \rangle|.$$

In a real Hilbert space, the states in the definition of $\mathcal{A}_\nu(H)$ reduce to pairs (x, x) , so $T \in \mathcal{A}_\nu(H)$ with $\nu(T) = 1$ means: for every $\varepsilon > 0$ there is $\eta(\varepsilon, T) > 0$ such that whenever $x \in S_H$ and

$$|\langle Tx, x \rangle| > 1 - \eta(\varepsilon, T),$$

there is $x_1 \in S_H$ with $\langle Tx_1, x_1 \rangle = 1$ and $\|x_1 - x\| < \varepsilon$.

Call a family $\mathcal{F} \subset \mathcal{L}(H)$ *numerically norming for H* if

$$\|x\|^2 = \sup_{T \in \mathcal{F}} |\langle Tx, x \rangle| \quad (x \in H).$$

Idea of the Proof

Proposition 2.8 uses a finite-dimensional identity block and an infinite compact skew-symmetric tail. The skew-symmetric part does not contribute to $\langle Tx, x \rangle$ over a real Hilbert space; only the identity block contributes. Thus if the identity block is F , then

$$\langle Tx, x \rangle = \|P_F x\|^2.$$

Near numerical-radius attainment forces x close to F , and normalizing $P_F x$ gives an exact numerical-radius attaining point. Since we are free to choose F , taking $F = \text{span}\{x\}$ gives a member of the family that norms any prescribed vector x .

Theorem

Theorem 1. *Let H be a separable infinite-dimensional real Hilbert space. There is a family $\mathcal{F} \subset \mathcal{L}(H)$ such that:*

1. *every $T \in \mathcal{F}$ is compact;*
2. *$1 = \nu(T) < \|T\|$ for every $T \in \mathcal{F}$;*
3. *every $T \in \mathcal{F}$ belongs to $\mathcal{A}_\nu(H)$ with the same admissible modulus $\eta(\varepsilon) = \varepsilon^2/4$ for $0 < \varepsilon < 1$;*
4. *$\mathcal{F}$ is numerically norming for H , that is,*

$$\|x\|^2 = \sup_{T \in \mathcal{F}} |\langle Tx, x \rangle| \quad (x \in H).$$

Indeed, the family can be chosen so that for every nonzero finite-dimensional subspace $F \subset H$, some $T_F \in \mathcal{F}$ satisfies $T_F|_F = I_F$.

Proof. Fix a nonzero finite-dimensional subspace $F \subset H$. Since H is separable and infinite-dimensional, choose an orthonormal decomposition

$$H = F \oplus \bigoplus_{k=1}^{\infty} \text{span}\{u_k, v_k\}.$$

Choose real scalars $\beta_1 > 1$, $0 < \beta_k < 1$ for $k \geq 2$, and $\beta_k \rightarrow 0$. Define $T_F \in \mathcal{L}(H)$ by

$$T_F f = f \quad (f \in F), \quad T_F u_k = -\beta_k v_k, \quad T_F v_k = \beta_k u_k.$$

The operator is compact: it is the direct sum of the identity on finite-dimensional F and 2×2 blocks whose norms β_k tend to zero. Also

$$\|T_F\| = \max\{1, \sup_k \beta_k\} = \beta_1 > 1.$$

Let $x = f + w$, where $f \in F$ and $w \in F^\perp$. The block operator on $F^\perp$ is skew-symmetric, hence

$$\langle T_F x, x \rangle = \|f\|^2 + \langle T_F w, w \rangle = \|f\|^2.$$

Therefore $\nu(T_F) \leq 1$, and equality holds on every unit vector of F . Thus $\nu(T_F) = 1$, and T_F attains its numerical radius.

We next verify the uniform $\mathcal{A}_\nu(H)$ estimate. Let $0 < \varepsilon < 1$, and suppose $x \in S_H$ satisfies

$$|\langle T_F x, x \rangle| > 1 - \varepsilon^2/4.$$

Writing $x = f + w$ as above, we get

$$\|f\|^2 > 1 - \varepsilon^2/4, \quad \|w\| < \varepsilon/2.$$

Set $x_1 = f/\|f\| \in S_F$. Then $\langle T_F x_1, x_1 \rangle = 1$, and

$$\|x - x_1\| \leq \|w\| + |1 - \|f\|| < \varepsilon/2 + \varepsilon^2/4 < \varepsilon.$$

Thus the same modulus $\eta(\varepsilon) = \varepsilon^2/4$ works for every T_F in the family.

Let $\mathcal{F} = \{T_F : F \subset H \text{ nonzero finite dimensional}\}$. For any $T_F \in \mathcal{F}$ and any $x \in H$,

$$|\langle T_F x, x \rangle| \leq \nu(T_F) \|x\|^2 = \|x\|^2.$$

Conversely, if $x \neq 0$, take $F = \text{span}\{x\}$. Then $T_F x = x$, so

$$|\langle T_F x, x \rangle| = \|x\|^2.$$

The equality is trivial for $x = 0$. Hence $\mathcal{F}$ is numerically norming for H . □

Verification and Limitations

The proof is a direct parameterization of the source construction, with the finite identity block allowed to vary. No computational verification is relevant.

This is recorded as a partial result because the source's phrase "norming for the whole space" is not formalized there. The theorem gives an affirmative answer for the standard numerical-vector interpretation:

$$\|x\|^2 = \sup_{T \in \mathcal{F}} |\langle T x, x \rangle|.$$

It does not address a possible interpretation in which the set of operators is meant to be norming for the Banach space $\mathcal{L}(H)$ in a dual-pairing sense.

Bounded novelty check: the run indexes had no exact packet for arXiv:1910.05726. Web searches on June 18, 2026 for the source title, the phrase "We do not know how often this happens", and related phrases involving $\mathcal{A}_\nu$, numerical radius, and norming did not reveal a later answer. The arXiv abstract page for 1910.05726 was used to confirm source metadata.

References

- [1] S. Dantas, M. Jung, and O. Roldan, *Norm attaining operators which satisfy a Bollobas type theorem*, arXiv:1910.05726.
- [2] F. F. Bonsall and J. Duncan, *Numerical Ranges of Operators on Normed Spaces and of Elements of Normed Algebras*, London Mathematical Society Lecture Note Series 2, Cambridge University Press, 1971.